Igor Uchoa

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Doctorate student at the Atmospheric and Oceanic Science Department of the University of Maryland. Master of Science in Physical Oceanography at the Oceanographic Institute at the University of São Paulo, Brazil. Currently, Doctorate Student at the University of Maryland (UMD) at the Atmospheric and Oceanic Science Department. Experienced in data analysis of oceanographic observational data and numerical simulations.

Education

Academic Qualifications

- **PhD Student Atmospheric and Oceanic Science** — **2021 - Present**
  University of Maryland, College Park, Maryland
- **M.Sc. Physical Oceanography** — **2017 - 2019**
  Oceanographic Institute of University of São Paulo, São Paulo, Brazil
- **B.S. Oceanography** — **2012 - 2016**
  Federal University of Ceará, Fortaleza, Brazil

Research

- **Graduate Assistant** — **College Park, MD**
  University of Maryland, College Park — *Aug 2021 - Present*
  Advisor: Jacob Wenegrat
- **Oceanographic Cruise Coordinator Assistant** — **São Paulo, Brazil**
  University of São Paulo — *7-21, Dez 2019*
  Coordinator: Prof. Ilson C. A. Silveira
- **Petróleo Brasil (PETROBRAS) / University of São Paulo** — **São Paulo, Brazil**
  Position: Oceanographer, Coordinator: Prof. Ilson C. A. Silveira — *Jul 2019 – Jul 2020*
  Main Activities: Oceanographic and Meteorological data processing; Model outputs and reanalysis processing; Plotting and sketch illustration for Data Science; Scientific writing for reports, papers, and presentations.

Teaching

- **Instructor** — **University of Maryland College Park**
  Weather and Climate Laboratory — *Aug 2025 - May 2026*
- **Teacher Assistant** — **University of Maryland College Park**
  Weather and Climate — *Aug 2025 - May 2026*
  Instructor: Jeffrey Henrikson
- **Instructor** — **Remote**
  Climatematch - Neuromatch Academy — *Jul 2024*
  Coordinator: Nick Halper
- **Teaching Assistant** — **São Paulo, Brazil**
  University of São Paulo, Course: Dynamic Oceanography II — *Jul 2018 – Dez 2018*

Coordinator: Prof. Ilson C. A. Silveira

- **Wise Up ESL School** — **Fortaleza/São Paulo, Brazil**
  Position: ESL Instructor — *Jan 2015 – Jul 2017*

Publications

Papers

- L. Renault, C. Conejero, J. Wenegrat, and I. **Uchoa**. Interplay of submesoscale current and thermal feedbacks: A seasonal perspective in the gulf stream. *Geophysical Research Letters*, 52(21), 2025

- I. **Uchoa**, J. Wenegrat, and L. Renault. Sink of eddy energy by submesoscale sea surface temperature variability in a coupled regional model. *Journal of Physical Oceanography*, 55, 2025

- J. T. Farrar and **et al.** S-MODE: The sub-mesoscale ocean dynamics experiment. *Bulletin of the American Meteorological Society*, 2025

- L. Renault, M. Contreras, P. Marchesiello, C. Conejero, **I. Uchoa**, and J. Wenegrat. Unraveling the impacts of submesoscale thermal and current feedbacks on the low-level winds and oceanic submesoscale currents. *Journal of Physical Oceanography*, 54(12):2463–2486, 2024

- L. Nuijens and **et al.** The Air-Sea Interaction (ASI) submesoscale: Physics and impact. In *Lorentz Workshop*, 2024

- F. Santos, L. Araújo, **I. Uchoa**, R. Lourenço, S. Taniguchi, C. Martins, R. Nagai, I. Wainer, M. Mahiques, R. Figueira, and M. Bícego. High-resolution UK'37 sea surface temperature reconstruction in the southwestern Atlantic Ocean during the Holocene. *Palaeogeography, Palaeoclimatology, Palaeoecology*, 654:112452, 2024

- **I. Uchoa**, I. T. Simoes-Sousa, and I. C. da Silveira. The Brazil Current mesoscale eddies: Altimetry-based characterization and tracking. *Deep Sea Research Part I: Oceanographic Research Papers*, 192:103947, 2023

- A. C. T. Bonecker, M. Katsuragawa, M. S. de Castro, C. Namiki, M. de Lourdes Zani-Teixeira, **I. Uchoa Farias**, and I. C. A. d. Silveira. Seasonal variability of ichthyoneuston assemblage on the continental shelf and slope of the Southwest Atlantic Ocean, Brazil (20–23 S). *Journal of applied ichthyology*, 35(3):655–671, 2019

Book Chapter

- I. C. A. da Silveira, D. C. Napolitano, and **Farias Uchoa, I.** *Water Masses and Oceanic Circulation of the Brazilian Continental Margin and Adjacent Abyssal Plain*, pages 7–36. Springer International Publishing, Cham, 2020

Advising

- **Co-advisor**
  Megan Brown - Undergraduate Research Project — *2024*

Guest Lecture

- Introduction to Oceanography - Weather and Climate (AOSC 200) - April 2026

Invited Talks

- **Title: Observed air-sea thermal feedback at submesoscale fronts**
  American Geophysical Union Annual Meeting, Washington DC *2024*
- **Title: Observed rapid adjustment of the atmospheric boundary layer to submesoscale sea surface temperature fronts**
  OASIS Air-Sea Webinar, Virtual *2026*

Service and Leadership

- **2024-2025** Letters for a Pre-scientist (LPS) - Volunteer
- **2024-Present** Communications for Science Policy Group - University of Maryland, College Park
- **2023-2024** Student Seminar Co-organizer - Atmospheric and Oceanic Science Department, University of Maryland - College Park

Additional Training

- **2022-2023** - Life Skills for Young Scientists, Planetary Sceince Institute

Achievements

- **2025** - Richard Payne Gratuate Award - University of Maryland, College Park
- **2024** - Green Fund Scholarship (Green Fund Endowed Graduate Fellowships in Earth Sciences and Climate Change) - University of Maryland, College Park
- **2024** - Group Achievement Award Submesoscale Ocean Dynamics Experiment (S-MODE) - NASA
- **2021** - College of Computer, Mathematical, and Natural Sciences Deans Fellowship - University of Maryland, College Park
- **2013** - Science Without Borders Scholarship - Brazil Government